SYNTHESIZED DESIGN - xc7z030ffg676-1

Tcl Console | Messages | Log | Reports | Design Runs | Utilization

```
open_run: Time (s): cpu = 00:00:20 ; elapsed = 00:00:13 . Memory (MB): peak = 2496.855 ; gain = 819.516
report_utilization -hierarchical_min_primitive_count 50 -name utilization_1
report_utilization
Copyright 1986-2022 Xilinx, Inc. All Rights Reserved. Copyright 2022-2025 Advanced Micro Devices, Inc. All Rights Reserved.
---------------------------------------------------------------------------------------------------------------------------------------------
| Tool Version : Vivado v.2025.2 (win64) Build 6299465 Fri Nov 14 19:35:11 GMT 2025
| Date         : Sat Apr 25 14:13:38 2026
| Host         : LAPTOP-SU6IC0U7 running 64-bit major release  (build 9200)
| Command      : report_utilization
| Design       : top_wrapper
| Device       : xc7z030ffg676-1
| Speed File   : -1
| Design State : Synthesized
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Utilization Design Information

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```

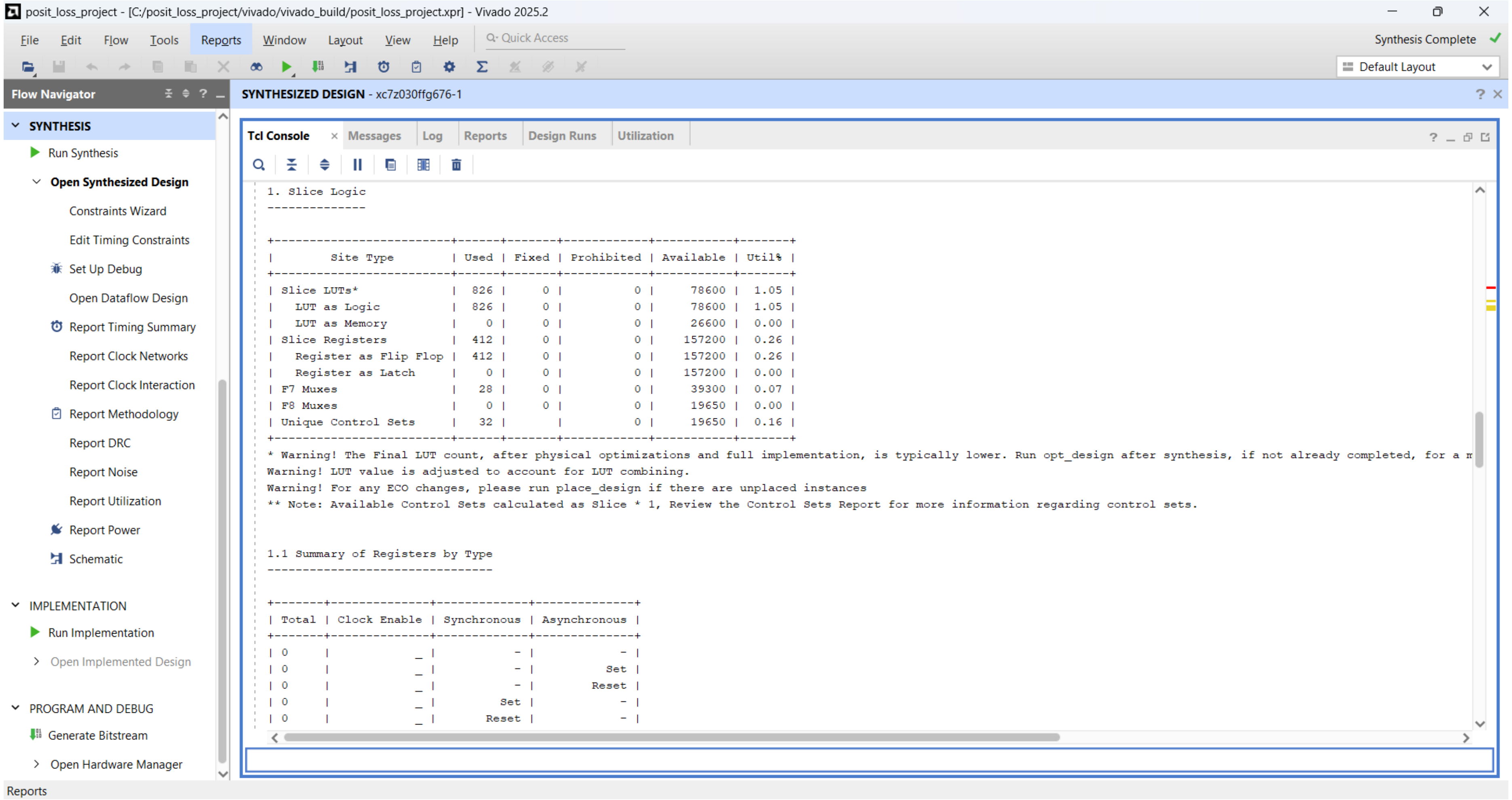
posit_loss_project - [C:/posit_loss_project/vivado/vivado_build/posit_loss_project.xpr] - Vivado 2025.2
File Edit Flow Tools Reports Window Layout View Help
Quick Access
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Flow Navigator
SYNTHESIZED DESIGN - xc7z030ffg676-1
SYNTHESIS
Run Synthesis
Open Synthesized Design
Constraints Wizard
Edit Timing Constraints
Set Up Debug
Open Dataflow Design
Report Timing Summary
Report Clock Networks
Report Clock Interaction
Report Methodology
Report DRC
Report Noise
Report Utilization
Report Power
Schematic
IMPLEMENTATION
Run Implementation
Open Implemented Design
PROGRAM AND DEBUG
Generate Bitstream
Open Hardware Manager
Tcl Console
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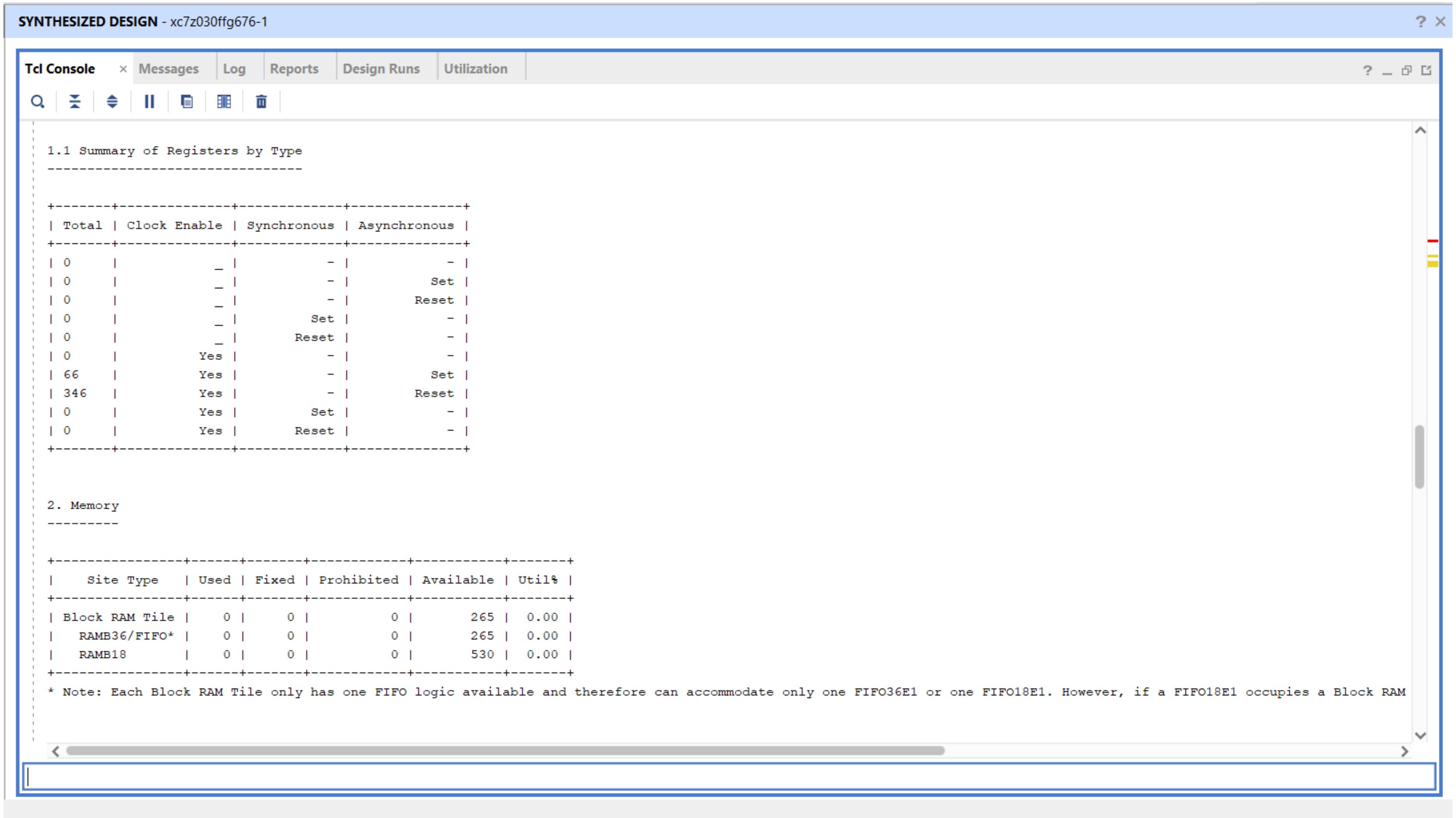
Utilization
1. Slice Logic
Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs*	826	0	0	78600	1.05
LUT as Logic	826	0	0	78600	1.05
LUT as Memory	0	0	0	26600	0.00
Slice Registers	412	0	0	157200	0.26
Register as Flip Flop	412	0	0	157200	0.26
Register as Latch	0	0	0	157200	0.00
F7 Muxes	28	0	0	39300	0.07
F8 Muxes	0	0	0	19650	0.00
Unique Control Sets	32		0	19650	0.16
* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a m
Warning! LUT value is adjusted to account for LUT combining.
Warning! For any ECO changes, please run place_design if there are unplaced instances
** Note: Available Control Sets calculated as Slice * 1, Review the Control Sets Report for more information regarding control sets.
1.1 Summary of Registers by Type
Total	Clock Enable	Synchronous	Asynchronous
0	_	-	-
0	_	-	Set
0	_	-	Reset
0	_	Set	-
0	_	Reset	-
Reports




```
1.1 Summary of Registers by Type
--------------------------------

+-------+--------------+-------------+--------------+
| Total | Clock Enable | Synchronous | Asynchronous |
+-------+--------------+-------------+--------------+
| 0     |            _ |           - |            - |
| 0     |            _ |           - |          Set |
| 0     |            _ |           - |        Reset |
| 0     |            _ |         Set |            - |
| 0     |            _ |       Reset |            - |
| 0     |          Yes |           - |            - |
| 66    |          Yes |           - |          Set |
| 346   |          Yes |           - |        Reset |
| 0     |          Yes |         Set |            - |
| 0     |          Yes |       Reset |            - |
+-------+--------------+-------------+--------------+


2. Memory
---------

+----------------+------+-------+------------+-----------+-------+
|    Site Type   | Used | Fixed | Prohibited | Available | Util% |
+----------------+------+-------+------------+-----------+-------+
| Block RAM Tile |    0 |     0 |          0 |       265 |  0.00 |
|   RAMB36/FIFO* |    0 |     0 |          0 |       265 |  0.00 |
|   RAMB18       |    0 |     0 |          0 |       530 |  0.00 |
+----------------+------+-------+------------+-----------+-------+
* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM
```

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3. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	1	0	0	400	0.25
DSP48E1 only	1				

4. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	65	0	0	250	26.00
Bonded IPADs	0	0	0	14	0.00
Bonded OPADs	0	0	0	8	0.00
Bonded IOPADs	0	0	0	130	0.00
PHY_CONTROL	0	0	0	5	0.00
PHASER_REF	0	0	0	5	0.00
OUT_FIFO	0	0	0	20	0.00
IN_FIFO	0	0	0	20	0.00
IDELAYCTRL	0	0	0	5	0.00
IBUFDS	0	0	0	240	0.00
GTXE2_COMMON	0	0	0	1	0.00
GTXE2_CHANNEL	0	0	0	4	0.00
PHASER_OUT/PHASER_OUT_PHY	0	0	0	20	0.00
PHASER_IN/PHASER_IN_PHY	0	0	0	20	0.00
IDELAYE2/IDELAYE2_FINEDELAY	0	0	0	250	0.00
ODELAYE2/ODELAYE2_FINEDELAY	0	0	0	150	0.00

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Tcl Console | Messages | Log | Reports | Design Runs | Utilization

5. Clocking

Site Type	Used	Fixed	Prohibited	Available	Util%
BUFGCTRL	1	0	0	32	3.13
BUFIO	0	0	0	20	0.00
MMCME2_ADV	0	0	0	5	0.00
PLLE2_ADV	0	0	0	5	0.00
BUFMRCE	0	0	0	10	0.00
BUFHCE	0	0	0	96	0.00
BUFR	0	0	0	20	0.00

6. Specific Feature

Site Type	Used	Fixed	Prohibited	Available	Util%
BSCANE2	0	0	0	4	0.00
CAPTUREE2	0	0	0	1	0.00
DNA_PORT	0	0	0	1	0.00
EFUSE_USR	0	0	0	1	0.00
FRAME_ECCE2	0	0	0	1	0.00
ICAPE2	0	0	0	2	0.00
PCIE_2_1	0	0	0	1	0.00
STARTUPE2	0	0	0	1	0.00

SYNTHESIZED DESIGN - xc7z030ffg676-1

Tcl Console | Messages | Log | Reports | Design Runs | Utilization

```
7. Primitives
-------------

+----------+------+---------------------+
| Ref Name | Used | Functional Category |
+----------+------+---------------------+
| LUT6     |  416 |                 LUT |
| FDCE     |  346 |        Flop & Latch |
| LUT3     |  152 |                 LUT |
| LUT5     |  149 |                 LUT |
| LUT2     |  116 |                 LUT |
| LUT4     |   75 |                 LUT |
| LUT1     |   69 |                 LUT |
| CARRY4   |   69 |          CarryLogic |
| FDPE     |   66 |        Flop & Latch |
| OBUF     |   36 |                  IO |
| IBUF     |   29 |                  IO |
| MUXF7    |   28 |               MuxFx |
| DSP48E1  |    1 |    Block Arithmetic |
| BUFG     |    1 |               Clock |
+----------+------+---------------------+


8. Black Boxes
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+----------+------+
| Ref Name | Used |
+----------+------+


9. Instantiated Netlists
```
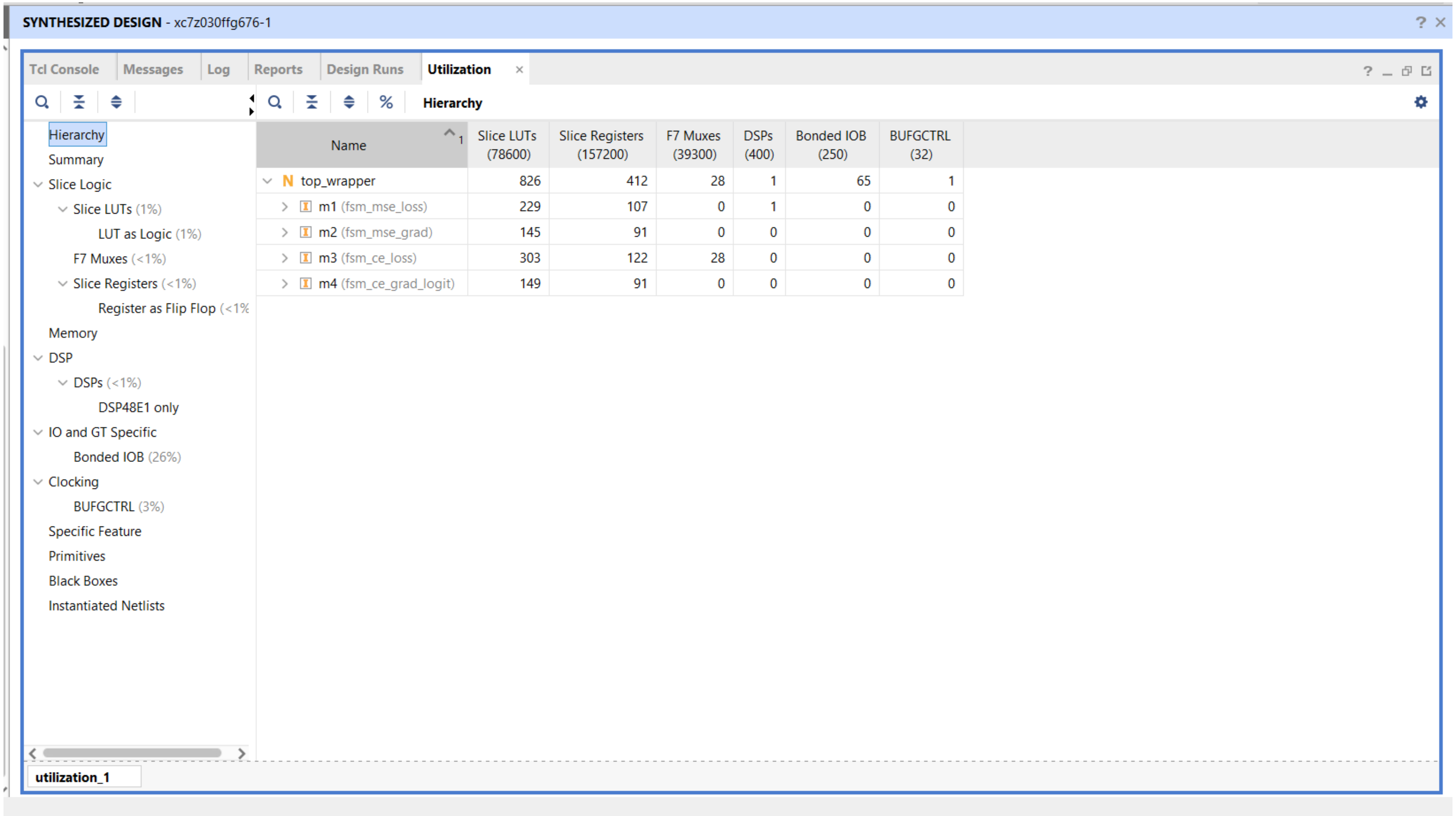
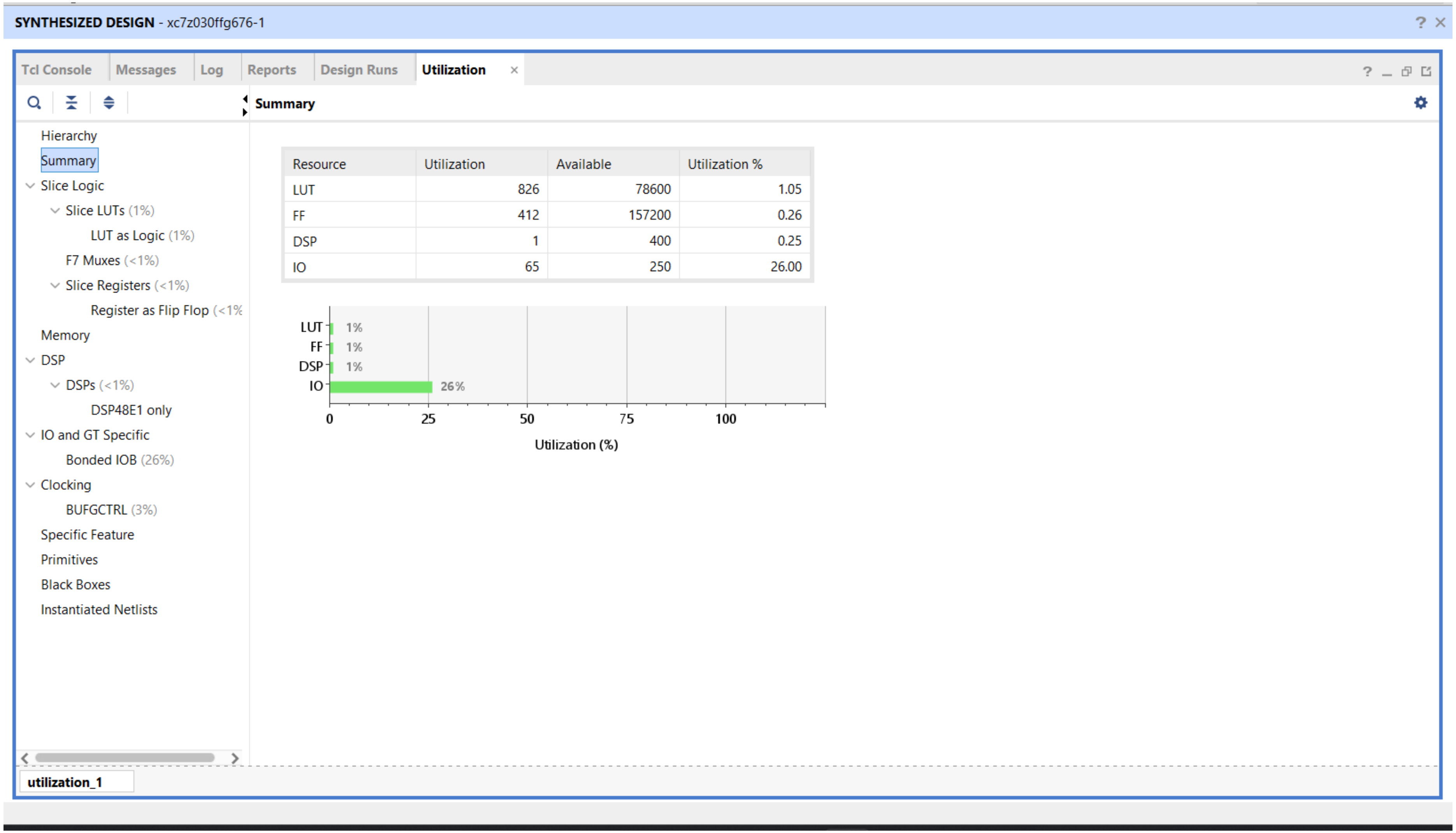



Name	Slice LUTs (78600)	Slice Registers (157200)	F7 Muxes (39300)	DSPs (400)	Bonded IOB (250)	BUFGCTRL (32)
top_wrapper	826	412	28	1	65	1
m1 (fsm_mse_loss)	229	107	0	1	0	0
m2 (fsm_mse_grad)	145	91	0	0	0	0
m3 (fsm_ce_loss)	303	122	28	0	0	0
m4 (fsm_ce_grad_logit)	149	91	0	0	0	0


SYNTHESIZED DESIGN - xc7z030ffg676-1
Tcl Console
Messages
Log
Reports
Design Runs
Utilization
Summary
Hierarchy
Summary
Slice Logic
Slice LUTs (1%)
LUT as Logic (1%)
F7 Muxes (<1%)
Slice Registers (<1%)
Register as Flip Flop (<1%
Memory
DSP
DSPs (<1%)
DSP48E1 only
IO and GT Specific
Bonded IOB (26%)
Clocking
BUFGCTRL (3%)
Specific Feature
Primitives
Black Boxes
Instantiated Netlists

Resource	Utilization	Available	Utilization %
LUT	826	78600	1.05
FF	412	157200	0.26
DSP	1	400	0.25
IO	65	250	26.00

LUT 1%
FF 1%
DSP 1%
IO 26%
0 25 50 75 100
Utilization (%)
utilization_1